

Manual

Weighing of Components AIP-KEW 8.1

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1 Weighing Components

Usage

For operations that are not subject to batch management, it is possible to record batches in relation to discrete material components. In this special case, batches may be entered in relation to the charge via a special terminal function.

This entry function replaces the collection of quantities for the operation (e.g. partial upload) and records material consumption in relation to material components. This consumption is also posted as material movement in the system.

Prerequisites

- The function can weigh more than one charge. In this case, the operation's secondary quantity has to include the number of charges. The additional data fields of the components and operation regarding the weighing function (e.g. target quantity per charge, tolerances, etc.) also need to be taken into account.
- Every time an operation is changed, the target quantity for each charge is recalculated (formula: $\text{target quantity per charge} = \text{primary target quantity} / \text{secondary target quantity}$). It is neither possible to set the value for the target quantity per charge manually nor to display it on MOC. If the secondary target quantity is not set, the value 1 is assumed by default and only one charge is processed in the weighing operation.
- In this case, the machine does not allow to enter automatic quantities additionally.
- All components have to be managed by the "weight" unit (kilogram).



The entry function cannot be used offline.

Configuration

The configuration document describes the configurations required to use and enable the entry function at the terminal.

Posting

Data for label printing

The schema **a_vbrkomp** is provided in initial data for label printing.

This data can be printed on the label during weighing (dialog action KEW_RECORD):

Acronym	Type	Length	Description
DLG.MNR	C	10	Machine
DLG.ANR	C	40	Operation
DLG.ATK	C	40	Material number
DLG.SLP	C	40	BOM item
DLG.CHARGE	C	20	Batch
DLG.CST	C	1	Batch status F/S (free/blocked)
DLG.SGR:GUT	DEC		Target quantity
DLG.SGE:GUT	C	3	Unit of the target quantity
DLG.EGR:GUT	DEC		Actual quantity
DLG.EGE:GUT	C	3	Unit of the actual quantity
DLG.EGR:MENGE	DEC		Input quantity
DLG.EGE:MENGE	C	3	Unit of the input quantity
DLG.DAT	DATE		Date of the posting
DLG.ZEI	ZEIT		Time of the posting
DLG.KNR	C	10	The reporting person's badge number

This data can be printed on the label when completing the charge (dialog action KEW_ABSCHLUSS):

Acronym	Type	Length	Description
DLG.MNR	C	10	Machine
DLG.ANR	C	40	Operation
DLG.ATK	C	40	Material number
DLG.ATKBEZ	C	40	Material designation
DLG.CHARGE	C	20	Batch
DLG.EGR:GUT	DEC		Quantity of the batch
DLG.EGE:GUT	C	3	Unit
DLG.DAT	DATE		Date of the posting
DLG.ZEI	ZEIT		Time of the posting
DLG.KNR	C	10	The reporting person's badge number

Scale interfacing

The weight can also be entered in the weighing dialog by means of a connected scale. Upon opening the KOMP_WIEG dialog, the value is requested from the scale via the PCC driver interface and entered in the "input quantity" field.

This is an example for entering the scale value in the INI file of the driver:

<WAAGENTREIBER>.INI
V:WAAGE:NETTO=Nettogewicht_Waage

If an OPC interfacing is used, changed scale values can be sent automatically by the OPC server. For this purpose, the below-mentioned parameter has to be entered in the file "OPCMPDV.INI"
SETVALEVENTS=V:WAAGE:NETTO

<WAAGENTREIBER>.INI
SETVALEVENTS=V:WAAGE:NETTO ← additional entry V:WAAGE:NETTO=Nettogewicht_Waage

2 Weighing of Components

Summary

Some production areas require components to be weighed using scales and to provide them for further production processes.

Usage

The function is always used when the user prepares charges using the relevant material quantities. To save time, users often weigh several charges at once.

The logical process and posting are described [here](#).

Prerequisite/configuration

The configuration is described [here](#).

Weighing components

Components for a charge are weighed by the AIP dialog A_VBRKOMP.



If the terminal is offline, only a limited number of posting functions is provided based on the available data. If the terminal is offline, errors are not displayed e.g. if posting failed on the server.

The "weighing" function at the terminal provides a list of required input materials (discrete material components). The component-related data collection can be opened from this list.

Once the operation to be weighed has been logged on, the "weighing components" dialog can be started by the "weigh" function key from the toolbar of the basic screen of the terminal. After opening the dialog, the terminal automatically generates a new batch for the order. This is presented as batch in the following dialog.

Please note: The dialog only opens if "discrete" material components (consumption type = D) exist for the running operation.

The actual weighing process of the relevant components is performed by the "weigh charge" function in a detailed dialog (see below).

Dialog

AIP
<A_VBRKOMP>

Workplace W2030
Operation M30002000010
Batch PR4RH6B11K

Material 4529-42
Designation

Status
☒ Free
☐ blocked

Components

POS	Article	Target qty.	Actual qty.	Remaining qty.	UOM	Designation	Tolerance	Dev
1	9901	169,000	0,000	0,000	L	Ethylalkohol	0,000	
2	3011	3380,000	0,000	0,000	G	Phosphorsäure	0,000	
3	4011	152,100	0,000	0,000	G	Glyzerin	0,000	
4	8511	16731,000	0,000	0,000	L	Destilliertes Wasser (Aquadest)	0,000	

Staff Number

Cancel
Weigh comp.
Quantity adjustm.
Reject charge
Complete charge

04.07.14 10:37:22

Workplace

Weighing workplace to which the operation to be weighed is logged on.

Operation

Weighing order/operation

Batch

Batch number for the charge that is currently to be produced.

Material

Material number for the charge that is currently to be produced.

Status

Batch status for the charge that is currently to be produced after weighing. This can be:

- Free
- Locked

Staff badge number

The user's staff badge number.

Components list

The component list shows all components of the currently registered weighing operation that are relevant for the charge to be produced. These details for the component are displayed to perform weighing:

- Article number (material number of the component from the component list)
- Target quantity
- Actual quantity (quantity entered upon weighing, at first 0,000)
- Remaining quantity (computed remaining quantity after weighing, at first 0,000)
- Designation (component name from the component list)
- Tolerance (calculated, admissible tolerance of the component/component list)
- Deviation (calculated, admissible deviation of the component/component list)

Procedure:

The user selects the first component that is to be weighed and starts weighing.

- The relevant component row is highlighted in **green** if the component is within the variance tolerance (at the component, percentage).
- Materials falling short of the variance tolerance are highlighted in **red**.
- Materials the weighing result of which exceeds the variance tolerance but is still within the tolerances of quantity adjustment (of the component) are highlighted in **blue**.
- The row is shown in black font, provided that the component has not yet been weighed.
- If the weight value is beyond the variance tolerance but within the tolerance of quantity adjustment, posting will be accepted but the user has to adjust quantities for the other components (automatically in the system). The input dialog cannot be closed if this modification is not made.

Quantities have to be entered for all input materials included in the component list.

Functions

The paragraphs that follow describe the functions provided in the input dialog to weigh components:

"Weigh charge"

The "weigh charge" function opens the dialog to weigh a selected individual component (KOMP_WIEG).

Weigh charge
<KOMP_WIEG>

Workplace	W2030	Batch	PR4RH6P113
Operation	M30002000010	Component	4529-42
		BOM item	1

Component batch

Default quantity 169,000 L

Actual qty. 150,000 KG

Balance 0,000 KG

Input quantity

Allowed deviation 0,000 %

Actual deviation 100,000 %

Staff Number

Cancel
Weigh charge

18.07.14 09:09:15

The current status of target and actual quantity is presented after selecting a component that is to be weighed. The user has to enter a relevant batch for consumption posting in the "component batch" input field. The batch has to be available in HYDRA with the appropriate material and the status "free" and it must have a remaining quantity ≥ 0 . The dialog can be closed by entering the weighing quantity and the staff badge number (optional).

Successful weighing has the following additional effects on the system:

- Indicators are recorded for the component
- The weighed quantity is added to the actual quantity of the component
- Material movements are generated for consumption (261)
- A batch assignment is generated for the batch ⇔ component batch.



- The article of the batch always has to match the article of the component.
- A component may also be weighed several times.
- The amount (stock) of component batches is only reduced if the relevant material type is assigned the "retrograde inventory collection" flag.

Once the weight has been entered, the displayed actual quantity and remaining quantity are updated.

The displayed component requirements are updated in the table once the component has been weighed successfully.

Weighing components

<A_VBRKOMP>

Workplace	W2030	Material	4529-42
Operation	M30002000010	Designation	
Batch	PR4RH6B11K		

Status
☒ Free
☐ blocked

Components

POS	Article	Target qty.	Actual qty.	Remaining qty.	UOM	Designation	Tolerance	Dev.
1	9901	169,000	150,000	19,000	L	Ethylalkohol	0,000	
2	3011	3380,000	150,000	3230,000	G	Phosphorsäure	0,000	
3	4011	152,100	0,000	0,000	G	Glyzerin	0,000	
4	8511	16731,000	0,000	0,000	L	Destilliertes Wasser (Aquadest)	0,000	

Staff Number 9999

Cancel
Weigh comp.
Quantity adjustm.
Reject charge
Complete charge

04.07.14 10:39:51

A warning message the user may skip is output if the value of the actual deviation is greater than the one of the target deviation.

"Reject charge" function

Due to the weighing process, it might be the case that an entire batch cannot be used. The "reject charge" function can be used to identify the current charge/batch as "scrap". The material used for this charge is uploaded to ERP and the generated scrap quantity (identified as charge scrap by SYSTEM reason 910) is posted to the order. The batch is generated with the "locked" status and the "scrap" batch class (goods movement 531).

A confirmation prompt the user has to affirm comes up with this function.

Then the original target quantity for each charge is restored and the components are reset. Consequently, the weighing process can be restarted for a new batch.

"Adjust quantity" function

Once the first component has been weighed, the user can apply the "adjust quantity" function to automatically adjust the input quantities of the other components in proportion to the weighing result of the first component and its default quantity. This function can only be used once for each charge.

The function adjusts the target quantities of all components to the weighed quantity of one component. Quantities are adjusted by changing the primary target quantity for each charge of the operation.

However, quantities may only be adjusted if the component is weighed within its tolerances.

"Complete charge" function

Using this function, the charge/batch is completed (generated) and the dialog is closed. However, the charge/batch can only be completed, once all components have been weighed and are within their specified tolerances.

Completed successfully, the charge/batch is generated and transferred as goods movement 101 to ERP. Optionally, the charge/batch is assigned a minimum shelf-life (from the material type of the OP). In addition, the PPS batch from the order is determined and stored as PPS batch in the batch of the charge (optionally).

The quantity of the generated batch results from the input quantity recorded as consumption for each component.

The user may also directly block the batch/charge by selecting a status from the component requirements dialog. By default, the batch is always assigned the status "free".

3 Configuration of Weighing Components

Usage

You use the "pass batch attributes on" function if attributes of input batches are to be transferred to the generated output batch when changing output batches.

Dialog configuration

The input function is controlled by the dynamic dialogs A_VBRKOMP and KOMP_WIEG

Activation of the posting function at the terminal

Specific posting functions are enabled at AIP by an entry in the file *ctaipbut.ini*.

This is an example for the entry in ctwinbut.ini:

CTAIPBUT.INI
F1=A_VBRKOMP, weigh



The dynamic dialogs A_VBRKOMP and KOMP_WIEG must be available.

System configuration

Operation data

The following additional fields have be filled out for the operation using the PPS interface:

- No batch management requirement
- Target quantity per charge (calculated form primary target quantity / secondary quantity)(ab.soll_menge_ansatz)
- Number of charges (secondary quantity in pieces) - default = 1
- Batch

Data included in component list

These parameters have to be filled out for discrete material components using the PPS interface:

- Tolerance (in percent) → mlst_hy.mengen_tol
- Deviation (absolute value) → mlst_hy.mengen_abweichung

- Input quantity and unit of input quantity
- Component type must be D - discrete